SHAWN NUGUID

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SUMMARY OF QUALIFICATIONS

- Third-year Computer Engineering student at Toronto Metropolitan University with hands-on experience in embedded firmware, bare-metal systems, hardware–software integration, PCB validation, robotics control, and technical documentation
- Experienced in C/C++, Python, Verilog/SystemVerilog, SQL, and JavaScript/TypeScript for developing automation scripts, testbenches, REST APIs, validation tools, data processing workflows, and performance reporting
- Strong hands-on hardware background using oscilloscopes, multimeters, power supplies, KiCad, MATLAB, ModelSim, and Linux-based tools to debug electronics, validate system interfaces, and improve reliability across systems impacting 1,000+ users

TECHNICAL SKILLS

Languages: C/C++ | **Python** | Java | VHDL | **Verilog** | **SystemVerilog** | SQL | JavaScript | TypeScript | R

Tools: Quartus II | KiCad | MATLAB | ModelSim | Visual Studio | Linux | Git | **Bash/Shell** | Arduino | RPi

Hardware: Oscilloscope | **Multimeter** | Ammeter | Power Supplies | Routers | Switches

EXPERIENCE

Hardware Design and Validation Engineer — **Mar. 2025 – May 2025**
Pocket Clinic Inc. — *Toronto, ON*

- Automated RFID-based **I2C/SPI** validation using **Verilog/SystemVerilog** testbenches, improving protocol coverage, test consistency, and issue tracking by **95%** across repeated validation cycles
- Developed and debugged **C-based firmware** and **Python** testbenches for smart injector operations and component degradation testing, using **MATLAB** and **Excel** to analyze results, identify problems, and improve debugging efficiency by **50%**
- Validated PCB schematics/layouts in **KiCad** using ERC/DRC checks while researching **ESP32** and **ATmega32** design considerations to support signal integrity, modularity, and future hardware integration

Software Developer — **July 2025 – Sep. 2025**
I Challenge Diabetes Camp — *Toronto, ON*

- Built a **Supabase SQL** UPSERT pipeline with **JSONB staging** to upload, validate, and structure camper records, improving database flow, data accuracy, and sync speed by **80%**
- Built and refactored HTTP REST APIs using **Node.js**, **Express**, **JavaScript**, and **TypeScript** for cloud-hosted applications, enabling scalable, real-time access to camper health data for over **200 parents** during camp
- Developed a **TypeScript** + **Supabase** CSV staging frontend using Papa.parse, implementing real-time validation rules to reduce upload errors, support cleaner data releases, and eliminate server downtime through continuous updates in Linux environments

Intramural Tech and Program Lead — **May 2025 – May 2026**
TMU Intramural & Sport — *Toronto, ON*

- Increased resume screening speed by **70%** by developing an **AI-powered ATS** using **Python**, **Pandas**, and **scikit-learn**, improving candidate prioritization through automated ranking and structured evaluation criteria
- Cleaned and maintained an internal database of over **16,000 users** using **Python** and **Excel**, removing outdated accounts to improve data accuracy, retrieval speed, and workflow efficiency
- Built **Tableau** dashboards for sports analytics, helping visualize participation trends, program performance, and operational decision-making

EDUCATION

Bachelor of Computer Engineering | Toronto Metropolitan University (TMU) — **April 2028**

Course Work: Microprocessor Systems, Control Systems, Computer Organization and Architecture, Digital Systems, Electric Circuit Analysis, Algorithms and Data Structures